

HOMEWORK 2

Problem 1

Use the provided dataset "HW2_data.csv" to build a multiple linear regression model

$$y = X\beta + \sigma\epsilon,$$

Where X is an $n \times p$ matrix. The size of the sample is $n = 50$ and the number of predictor variables is $p = 3$. Note that the column of 1's for the intercept term is already included in the dataset.

You can use programming software (R, Python, etc.) to do the computations. You do not need to submit the code but please show your work, including equations, test statistics and conclusions.

- (1) (10 pts) Find the estimator $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)$ and the distribution of $\hat{\beta}_3$.
- (2) (10 pts) Perform an F -test for the hypothesis $H_0 : \beta_3 = 0$ with the $\alpha = 0.01$ level of significance. Provide the value of the F test statistic and the corresponding critical value.
- (3) (10 pts) Use the marginal distribution of $\hat{\beta}_3$ to conduct a t -test for the hypothesis $H_0 : \beta_3 = 0$ with the $\alpha = 0.01$ level of significance. Provide the value of the test statistic and the p value.
- (4) (10 pts) Compute the leverage score for the first and second observations. Which data point is more influential? What is the issue arise with a very high leverage value.
- (5) (10 pts) Given a new data point $x = (1, 0.12, 0.56)$, find the predicted y value and its 95% confidence interval.

Problem 2

No dataset will be used for this question. All answers should be in mathematical form. Consider the p dimensional multiple linear model

$$y = X\beta + \sigma\epsilon$$

where $X'X = D$ is a diagonal matrix, $D = \text{diag}(d_j)$, and one of these elements, say $d_1 = \eta$ is very small and $1/\eta$ is very large.

- (1) (10 pts) What is the problem with using the ordinary least squares estimate; i.e.

$$\hat{\beta} = (X'X)^{-1}X'y?$$

- (2) (10 pts) A ridge estimator is proposed; i.e.

$$\hat{\beta}_R = (X'X + \lambda I)^{-1}X'y.$$

How does this solve the problem alluded to in part (1). In particular, find the mathematical form of the variance of the ridge estimator for β_1 .

- (3) (10 pts) What is the bias of the ridge estimator; i.e. $E(\hat{\beta}_R) - \beta$.
- (4) (10 pts) When choosing a value for λ , why is it important not to set the value to be either too large or too small.
- (5) (10 pts) For what values of λ , in terms of η , σ and the true value of β_1 , is it that the MSE of $\hat{\beta}_{R1}$ is smaller than that of the corresponding value of $\hat{\beta}_1$.